# XIHAN DENG

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## Education

**Shanghai Jiao Tong University**
Bachelor of Engineering — Sep. 2023–Jun. 2027 (expected)

- IEEE Honor Class
- Major: Computer Science and Technology; Second Major: Mathematics and Applied Mathematics
- GPA: 4.07/4.30; Weighted Average Score: 93.8/100
- Earned A+ grades (95–100/100) in all courses for the mathematics second major to date

## Publications and Manuscripts

*Authors are listed alphabetically, following the convention in theoretical computer science.*

1. The Price of Anarchy for Selfish Load Balancing
   **Xihan Deng**, Yaonan Jin, Wenqian Wang, and Yuhao Zhang
   *67th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2026)*

2. The Price of Anarchy for Selfish Load Balancing under Correlated Priors
   **Xihan Deng**, Yaonan Jin, Wenqian Wang, and Yuhao Zhang
   *International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW 2026)*

3. Stratifying Markets for Efficient Trade
   **Xihan Deng**, Yifan Lin, Daniel Schoepflin, and Kangning Wang
   *Manuscript*

## Selected Honors

- China National Scholarship for Undergraduate Students — 2025
  - One of China's most prestigious undergraduate scholarships

## Research Experience

**DIMACS, Rutgers University**
Advisor: Prof. Kangning Wang — Apr. 2026–Present

- **Stratifying Markets for Efficient Trade**
  - Analyzed simple mechanisms for bilateral trade and established their limitations in achieving approximate efficiency.
  - Introduced a market stratification model that achieves approximate efficiency in bilateral trade and double auctions, and established its limitations in matching markets.

**John Hopcroft Center, Shanghai Jiao Tong University**
Advisors: Prof. Yuhao Zhang and Prof. Yaonan Jin — Jul. 2025–Present

- **The Price of Anarchy for Selfish Load Balancing**
  - Proved tight bounds on the Price of Anarchy for selfish load balancing under Bayesian Nash Equilibria with independent and correlated priors, as well as under Correlated and Coarse Correlated Equilibria.

## Teaching

- **Teaching Assistant** at Shanghai Jiao Tong University — Spring 2026
  *Algorithm Design and Analysis (AI2615)*

## Language Skills

- **Mandarin Chinese:** Native
- **English:** Fluent; TOEFL iBT: 5.5/6.0 (110/120, CEFR C1), with scores of at least 5.0/6.0 in every section